

Excited-state energy component analysis for molecules - Why the lowest excited state is not always the HOMO-LUMO transition

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Abstract

The ability to tune excited-state energies is crucial to many areas of molecular design. In many cases this is done based on the energies of the highest occupied (HOMO) and lowest unoccupied (LUMO) molecular orbitals. However, this viewpoint is incomplete neglecting the many-body nature of the underlying excited-state wavefunctions. Within this work we highlight the importance of two crucial terms, other than orbital energies, that contribute to the excitation energies and show how to quantify them from quantum chemistry computations: an attractive Coulomb and a repulsive exchange interaction. Using this framework, we explain under which circumstances the lowest excited state of a molecule, of either singlet or triplet multiplicity, is not accessed via the HOMO/LUMO transition and show two paradigmatic examples. In the case of the push-pull molecule **ACRFLCN** we highlight how the lowest triplet excited state is a locally excited state lying below the HOMO/LUMO charge transfer state due to enhanced Coulomb binding. In the case of the naphthalene molecule, we highlight how the HOMO/LUMO transition (the 1L_a state) becomes the second excited singlet

state due to its enhanced exchange repulsion term. More generally, we explain why excitation energies do not always behave like orbital energy gaps, providing insight into photophysical processes as well as methodological challenges in describing them.

Introduction

The ability to tune electronic excitation energies of molecules is crucial in the development of materials for optoelectronic applications, and particularly the relative energies of the first singlet and triplet excited states is a prominent parameter of high current interest.¹⁻³ The design of new molecules is usually based on the energies and shapes of the frontier molecular orbitals (HOMO and LUMO) of the molecules studied. This viewpoint provides a useful starting point, e.g., when estimating the expected effect of different substituents. However, the HOMO/LUMO picture does not tell the whole story^{4,5} and there are many molecules where the HOMO/LUMO transition does not constitute the lowest excited state of a given spin multiplicity. The HOMO and LUMO energies obtained can be strongly method dependent.^{6,7} More specifically, it is challenging to rationalise singlet-triplet gaps from viewing the HOMO and LUMO independently. Considering, thus, the wide ranging applications but also shortcoming of the HOMO/LUMO picture it could be highly beneficial to have a systematic way to go beyond it. In particular, the availability of a scheme for decomposing excitation energies into physically motivated energy components could have great potential.

For ground state molecules a variety of popular energy decomposition analysis schemes are available.⁸⁻¹⁰ These provide a quantitative decomposition of the computed interaction energies for bonding and non-covalent interactions into different physically motivated energy components, e.g., Coulomb, exchange and polarization, providing detailed insight into the interactions studied. By contrast, much of the focus in the analysis of excited state computations is on a qualitative characterization of state character, for example in terms of charge-transfer,¹¹⁻¹⁴ double excitations,^{15,16} and multiconfigurational character.¹⁷ The tun-

ing of excitation energies, on the other hand, usually proceeds via qualitative rules of thumb with no quantitative counterpart, e.g. regarding electron donating/withdrawing groups or delocalization and – for singlet/triplet gaps – in terms of HOMO/LUMO overlap,¹⁸ radical character¹⁹ or excited-state aromaticity.²⁰ More quantitative approaches to decomposing excitation energies^{21,22} have not reached maturity for routine use. This is arguably because the physical meaning of the terms involved is not *a priori* clear, no graphical representations are available, and the benefits of such methods to routine problems have not been highlighted.

It is the purpose of this work to present a clear physical picture for decomposing excitation energies into physically meaningful energy components combined with an intuitive graphical representation and to use this scheme to explain the breakdown of the MO picture. Using preliminary ideas by us and co-workers^{22–24} we develop a consistent and physically meaningful framework providing well-defined excitation energy components for quantum chemistry computations. Building on the MO picture, we provide a clear set of correction terms that bridge between the MO energy gap and the full excitation energy. The energy terms are readily connected with a graphical analysis scheme presented previously.²⁴ We use the developed tools to highlight under what circumstances the lowest excited state (of either singlet or triplet spin multiplicity) of a molecule is not derived from the HOMO/LUMO transition. More generally, we highlight the influence of an attractive Coulomb term and a repulsive exchange term on the state ordering and relative energies.

Two paradigmatic examples are chosen to illustrate the energy component analysis and, in particular, to highlight the breakdown of the HOMO/LUMO picture in systems of practical interest: the push-pull molecule **ACRFLCN** (10-phenyl-10H-spiro[acridine-9,9'-fluorene]-2',7'-dicyanitrile) and the naphthalene molecule (Fig. 1). The **ACRFLCN** molecule²⁵ is chosen to represent the class of donor-acceptor systems with low singlet-triplet gap employed in thermally activated delayed fluorescence (TADF). Within TADF emitters the interplay of different locally excited (LE) and charge transfer (CT) states is particularly important where CT character is decisive for lowering the singlet-triplet gap whereas LE character

promotes spin-orbit coupling.²⁶ Below we will investigate the different energy components determining the relative stability of these LE and CT states. Naphthalene, on the other hand, represents the case of a typical aromatic molecule where quasidegeneracies in the π -system lead to a set of $\pi\pi^*$ states of quite distinct character despite involving similar orbital transitions.²⁷⁻²⁹ In this case, we illustrate how our energy component analysis casts new light onto the differences between the ionic “+” and covalent “-” states present in these systems.^{30,31} We will explain the consequences on their spectroscopic properties as well as on their computational description with ab initio methods.^{32,33} Below, we will show that the lowest singlet state (S_1) of **ACRFLCN** is indeed the HOMO-LUMO transition whereas the triplet (T_1) is not. Naphthalene represents the opposite case where T_1 is dominated by the HOMO-LUMO transition but S_1 is not.

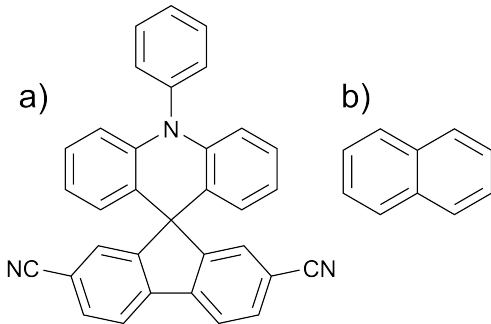


Figure 1: The molecular structures of (a) ACRFLCN and (b) naphthalene.

Methods

Within this section we start with a general discussion of a physically motivated energy decomposition proceeding from a two orbital model to general ab initio computations. We will continue with a more technical part outlining the detailed energy terms involved in range-separated hybrid time-dependent density functional theory (TDDFT). Subsequently, a more general interpretation of the significance of the derived energy terms is presented. We conclude with the computational details. Within this section, we aim for a compact and

intuitive explanation of the terms involved while a more detailed derivation is presented in Ref. 24.

Excited-state energy decomposition

We use a two-orbital two-electron model (TOTEM)^{24,34,35} as a conceptual starting point to illustrate the physics involved and proceed to a more general discussion later. The singlet and triplet excitation energies within the TOTEM are presented in Fig. 2. The first component, applicable to singlets and triplets alike, is the energy required to promote the electron from the occupied orbital ϕ_i to the virtual orbital ϕ_a . It is represented by the orbital energy gap $\epsilon_a - \epsilon_i$.

However, the orbital energy gap is not generally a good guess for the excitation energy. To explain this, we start with the simplest possible construction of an excited state, a triplet state formed as a single Slater determinant; an electron is promoted from ϕ_i and to ϕ_a with its spin flipped. To obtain the energy of this Slater determinant, we first have to remove the electron from ϕ_i . According to Koopmans' theorem this requires an energy of ϵ_i , given as

$$\epsilon_i = h_i + (ii|ii) \quad (1)$$

where h_i is the one-electron term and $(ii|ii)$ is the Coulomb repulsion between the two-electrons located in orbital ϕ_i . To reattach the electron, we can also apply Koopmans' theorem using the orbital energy of ϕ_a as an estimate of the electron affinity. However, the crucial realization is that we have to use the orbital energy evaluated with respect to the charged system

$$\epsilon_a^+ = h_a + (ii|aa) - (ia|ia), \quad (2)$$

rather than the orbital energy in the neutral system

$$\epsilon_a = h_a + 2(ii|aa) - (ia|ia) \quad (3)$$

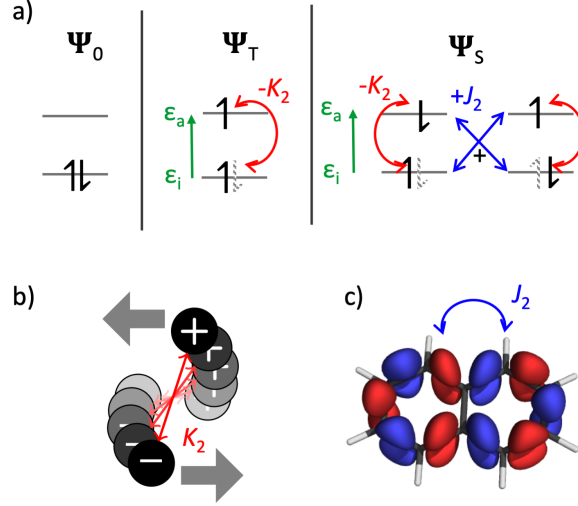


Figure 2: Depiction of energy terms contributing to excited-state energies: (a) excitation energies of triplet (Ψ_T) and singlet (Ψ_S) excited states within a two-orbital two-electron model; generalisation of (b) the K_2 term as a dynamic electron-hole binding energy, and (c) the J_2 term as a transition density repulsion.

which differ by one Coulomb term ($ii|aa$). In summary, the energy of a triplet state in the TOTEM is

$$\Delta E_T = \epsilon_a^+ - \epsilon_i = \epsilon_a - \epsilon_i - (ii|aa) \quad (4)$$

The energy is shifted by the Coulomb attraction ($ii|aa$) between occupied and virtual orbitals, a term that we will generally denote as K_2 . K_2 can be interpreted in the sense that the excited electron is attracted by the *hole* it left in the ground-state density (indicated as a dashed line in Fig. 2).

Open-shell singlet excited states are generally constructed from at least two Slater determinants and, hence, the derivation of the energy terms is slightly more involved;^{24,36} one obtains the expression

$$\Delta E_S = \epsilon_a - \epsilon_i - (ii|aa) + 2(ia|ia) \quad (5)$$

containing the K_2 term and an additional term $2(ia|ia)$ that we will generally denote J_2 . The J_2 term only applies to singlet states. It is generally repulsive and interpreted as an exchange interaction between electron and hole.

To turn the above consideration into a practically useful tool, it is worth realising that the energies of most excited-state quantum chemistry methods naturally comply with the above decomposition. Taking configuration interaction singles (CIS) or TDDFT in the Tamm-Dancoff approximation (TDA) as an example,^{6,24} the excitation energies can be written in the following way

$$\Delta E = \underbrace{\sum_{ia} |C_{ia}|^2 (\epsilon_a - \epsilon_i)}_{h'} + J_2 + K_2 + XC_2 . \quad (6)$$

Here the C_{ia} are the TDDFT response coefficients and J_2/K_2 are the two-electron terms deriving from the response of the Coulomb and non-local exchange potentials. The precise forms of these terms will be discussed below but for now it is enough to realise that they are natural generalisations of the $(ia|ia)$ and $(ii|aa)$ terms mentioned above. The XC_2 term contains any additional contributions deriving from exchange or correlation effects. Within TDDFT the XC_2 term is explicitly evaluated as the response of the exchange-correlation potential. Within CIS, the XC_2 term vanishes. However, it would be a natural generalisation within a correlated ab initio framework to define XC_2 as the energy difference between CIS and the correlated method used.

Furthermore, the one-electron term, denoted h' is given as a weighted sum of orbital energy differences as in Eq. (6). It can be written explicitly as

$$h' = \sum_{ia} |C_{ia}|^2 (\epsilon_a - \epsilon_i) = H + J_1 + K_1 + XC_1 \quad (7)$$

containing the one-electron core Hamiltonian (H) and one-electron terms J_1 , K_1 , XC_1 computed from the ground-state Fock matrix. However, we will not use this decomposition in the following but consider the h' term as a whole.

All terms in Eqs (6) and (7) can be readily obtained from CIS or hybrid TDDFT/TDA computations as reported previously^{22,23} providing an interesting practical basis for our

analysis.

Practical evaluation

Eq. (6) provides a well-defined decomposition of excitation energies in terms of the MO energy gaps and three additional two-electron terms. However, there is some arbitrariness in terms of how to compute the individual terms. From a physical point of view one can study why the excitation energy is not equivalent to a difference in electron affinity and ionisation potential and what other terms contribute. From a more applied point of view one can ask if, for any given method, MO energy gaps provide a good guess to the excitation energies. In this context it should be noted that orbital energies can be strongly method dependent differing between Hartree-Fock and the various density functionals. Within Hartree-Fock the HOMO/LUMO gap represents – at least in principle – the fundamental gap via Koopmans theorem. By contrast, for Kohn-Sham DFT with local functionals (e.g. BLYP, PBE) or global hybrid functionals with not too much exchange (e.g. B3LYP, PBE0) the HOMO/LUMO gap is already a good guess for the excitation energy noting that occupied and virtual orbital experience the same Kohn-Sham potential.^{6,7,37} Within the presented formalism, the difference is encoded in a balance between the K_1 and K_2 terms. Enhanced non-local exchange increases the HOMO/LUMO gap via an increased K_1 term while this is compensated for the excitation energy via an enhanced K_2 term. As a consequence, the excitation energies for locally excited states are fairly insensitive to the amount of non-local exchange despite dramatic changes in the HOMO/LUMO gap.

Within this work we use TDDFT with an optimally tuned range-separated density functional. This is aimed at combining both viewpoints just described, since the obtained HOMO and LUMO energies are *per construction* good guesses of the ionisation potential and electron affinity.³⁸

Within TDDFT/TDA and CIS, the J_2 term of Eq. (6) is evaluated as a linear combination

of exchange-type two-electron integrals

$$J_2 = 2 \sum_{abij} C_{ia} C_{jb} (ia|jb). \quad (8)$$

where C_{ia} are the TDDFT response coefficients (summed over α and β spins). To obtain a more intuitive form of this equation, it is first convenient to introduce the one-electron transition density matrix

$$\gamma_t(r_h, r_e) = 2 \sum_{ia} C_{ia} \phi_i(r_h) \phi_a(r_e) \quad (9)$$

which is interpreted as the wavefunction of the electron-hole pair, described by the coordinates r_e and r_h .^{14,24} Subsequently, the transition density (ρ_t) is defined as $\rho_t(r) = \gamma_t(r, r)$. Using this definition, Eq. (9) is rewritten as²⁴

$$J_2 = \iint \frac{\rho_t(r_1) \rho_t(r_2)}{r_{12}} dr_1 dr_2 = \langle \rho_t | \hat{V} | \rho_t \rangle \quad (10)$$

where r_{12} is the interelectronic separation. J_2 is simply the Coulomb repulsion of the transition density with respect to itself, as illustrated in Fig. 2 (c). As shown on the r.h.s. of Eq. (10), the J_2 term can be viewed as the overlap of the transition density (ρ_t) with the electrostatic potential (ESP) induced by it ($\hat{V} \rho_t$) and we suggest using this for a graphical representation (see below).²⁴

The K_2 term within range-separated TDA-TDDFT is slightly more involved. It can be written as a linear combination of modified two-electron integrals (cf. Refs 6,39)

$$K_2 = - \sum_{abij} C_{ia} C_{jb} \left(ij \left| \frac{s(r_{12})}{r_{12}} \right| ab \right) \quad (11)$$

where we use $s(r_{12})$ to denote the function used for the range separation procedure. Within the LRC- ω PBEh functional⁴⁰ employed here this function is defined as

$$s(r_{12}) = C_{\text{HF}} [1 - \text{erf}(\omega r_{12})] + \text{erf}(\omega r_{12}) \quad (12)$$

where C_{HF} is the global fraction of Hartree-Fock exchange, ω is the range-separation parameter, and erf denotes the error function. Note that for a global hybrid functional one would simply set $s(r_{12}) = C_{\text{HF}}$ and the analogous term for configuration interaction singles (CIS) is obtained with $s(r_{12}) = 1$.

Using Eq. (9), K_2 is re-expressed in the form

$$K_2 = - \iint |\gamma_t(r_h, r_e)|^2 \frac{s(r_{he})}{r_{he}} dr_h dr_e \quad (13)$$

Thus, K_2 can be seen as a screened Coulomb interaction between the electron and hole quasiparticles where the screening is determined by the function $s(r_{12})$.

Eq. (13) highlights that K_2 is generally a correlated term depending in a non-trivial fashion on the two-body function $\gamma_t(r_h, r_e)$ describing the mutual distribution of *electron* and *hole*; this dynamic nature is represented graphically in Fig. 2 (b). Conversely, if no correlation is present and γ_t can be factorized into a single pair of natural transition orbitals (NTOs), then K_2 takes on the simpler form of a (screened) Coulomb interaction between the *hole* (ρ_h) and *electron* (ρ_e) densities²⁴

$$K_2 \approx - \iint \rho_h(r_h) \rho_e(r_e) \frac{s(r_{he})}{r_{he}} dr_h dr_e \quad (14)$$

Under this approximation, the K_2 term can be represented via plots of ρ_h/ρ_e and their ESPs in analogy to Eq. (10). But it should be remembered that this works only if γ_t can indeed be represented by only two NTOs (see Refs 17,24 for further discussion).

The last term contributing to the TDA-TDDFT energy is XC_2 , the response of the exchange-correlation potential. It is given in analogy to Eq. (8) only that the electron repulsion integrals are replaced with two-electron integrals ($ia|f_{xc}|jb$) involving the response of the exchange-correlation potential⁶

$$XC_2 = \sum_{abij} C_{ia} C_{jb} (ia|f_{xc}|jb) \quad (15)$$

Table 1: Physical meaning of the four terms used in the energy decomposition analysis

Term	Physical interpretation
h'	Independent ionization and electron attachment
J_2	Exchange repulsion between electron and hole
K_2	Screened Coulomb binding between electron and hole
XC_2	Further exchange or correlation effects

There is no simple interpretation for the XC_2 term, it is simply a collection of all contributions to the excitation energy that are not related to the orbital gap, Coulomb attraction or exchange repulsion. As shown below, within the presented approach, XC_2 represents only a small correction term not affecting the qualitative discussions.

Interpretation of the energy terms

A summary of the four relevant terms is presented in Tab. 1. h' is the weighted MO energy difference and represents the energy required for independent ionization and electron attachment. This term is routinely invoked when explaining excited state energies based on MO energy gaps. Three two-electron terms follow describing the interactions between electron and hole.

J_2 signifies the exchange repulsion between the two electrons present in open-shell singlet states; it vanishes for triplet states (assuming a singlet ground state) considering that also the transition density ρ_t vanishes in space. J_2 is the dominant term inducing the splitting between singlet and triplet states. Its value is large if the *electron* and *hole* orbitals overlap in space. The exchange repulsion determines the magnitude of the gap between the first singlet and triplet excited states, a crucial quantity when designing chromophores for a number of applications, e.g., thermally activated fluorescence, singlet fission, or triplet upconversion.¹⁻³

The K_2 term corresponds to a binding energy of the electron-hole pair. It is generally a stabilizing term representing the dynamic Coulomb attraction between the *electron* and *hole* quasiparticles. This Coulomb attraction is the primary contribution to the exciton binding energy, that is, the energy required to separate the *electron-hole* pair into free charge

carriers, which plays an important role in the operation of solar cells.^{41,42} As explained in more detail above, K_2 is generally a correlated term depending on the two-body distribution of electron and hole. It is by virtue of this term that TDDFT and CIS can describe certain kinds of nondynamic correlation, present in dynamic exciton binding⁴³ and charge-resonance effects.⁴⁴

Contrasting J_2 and K_2 it is interesting to note that the response of the Coulomb potential J_2 acts as an exchange-like term on the *electron-hole* pair and *vice versa* for the K_2 term inducing Coulomb binding (see also Ref. 6). This induces some potential for confusion in terms of the definition of the terms; for the present purposes, we choose to use the J and K labels according to the ground-state operators rather than the *electron-hole* picture. Furthermore, the presented formalism suggests that J_2 should be computed with an unmodified or “bare” Coulomb interaction whereas K_2 uses a screened Coulomb interaction as defined by the range-separation procedure. Note that this distinction between bare exchange and screened Coulomb interaction is also common in the solid-state physics community where the latter is the eponymous W -term of the GW method.^{45,46}

From a technical point of view, it is worth noting that K_2 is the critical term in the TDDFT description of charge transfer states⁴⁷ and delocalized excitons⁴³ where the question of just how much the Coulomb interaction is screened (via C_{HF} and ω) is decisive on how well such states are described. Conversely, an accurate description of the J_2 term needs to include σ -correlation,²⁴ as discussed in more detail below. This fact underlies the long-known challenges^{48–50} of describing ionic $\pi\pi^*$ states with π -only correlation models.

Computational Details

Vertical excitations for naphthalene and **ACRFCLN** were computed using LRC- ω PBEh/def2-SV(P)^{40,51,52} with 20% global Hartree-Fock exchange ($C_{HF} = 0.20$). Optimally tuned values of $\omega = 0.140$ a.u. for **ACRFCLN** and $\omega = 0.225$ a.u. for naphthalene were used. Here, a tuning procedure was employed that matches the HOMO and LUMO energies of the neutral

system most closely to ionisation potential and electron affinity, respectively. The geometries were obtained at the same level of theory using the original parameters for LRC- ω PBEh ($\omega = 0.200$ a.u., $C_{\text{HF}} = 0.20$). No solvent model was employed.

Analyses were performed as described above. In addition, the exciton size d_{exc} defined as a root-mean-square electron-hole separation was computed:¹⁴

$$d_{exc}^2 = \Omega^{-1} \iint |\gamma_t(r_h, r_e)|^2 r_{he}^2 dr_h dr_e \quad (16)$$

where Ω is a normalisation factor whose value is 1 in the case of TDA-TDDFT. The d_{exc} value has been shown to be a versatile measure for charge transfer,¹⁴ exciton binding,⁴³ and Rydberg character.⁵³ We will study its relation to the K_2 term here noting the similarity of Eq. (16) to Eq. (13).

In addition, we compute a descriptor, LOC, measuring the probability that *electron* and *hole* are located on the same basis function simultaneously. It is computed as

$$\text{LOC} = \sum_{\mu} \tilde{D}_{\mu\mu}^2 \quad (17)$$

where the $\tilde{D}_{\mu\mu}$ are the diagonal elements of the transition density matrix expressed within the Löwdin-orthogonalised atomic orbital basis. LOC is related to the previously defined fragment-based CT value⁴⁴ where LOC is equivalent to 1-CT with each basis function seen as its own fragment. As shown below states with $LOC = 0.000$ are associated to “-” states in Pariser’s nomenclature whereas $LOC > 0.03$ indicates “+” states.

Calculations were performed using a developmental version of Q-Chem 5.2/6.0.⁵⁴ The libwfa library in Q-Chem is used for wavefunction analysis, ESPs, and for computing the LOC descriptor.⁵⁵ For technical reasons, ESPs were always computed with an unscreened Coulomb interaction [i.e. $s(r_{he}) = 1$ in Eq. (14)]. Excited state energy component analysis is performed by specifying `EXCIT_ENERGY_COMPONENTS = TRUE` in the Q-Chem input files.²² The TheoDOR toolkit is used for general post-processing of excited state

computations.⁵⁶ PyMOL is used with the external `qc_pymol` toolkit for plotting ESPs.^{57,58} Q-Chem input/output files are provided via a separate repository.⁵⁹

Results and discussion

Armed with the above toolkit we now want to tackle an intriguing question: When is the lowest state not the HOMO-LUMO transition? The first term in Eq. (6) is per construction minimised by the HOMO-LUMO transition, since it directly depends on the MO energies. Conversely, the three two-electron terms are responsible for the fact that excitation energies are not equal to orbital energy gaps and if they differ enough for different types of states, then the HOMO/LUMO transition can be pushed above another state. Two cases can be envisaged. If the stabilizing K_2 term of the HOMO-LUMO transition is small, then another state with larger K_2 may be moved below it. This case is particularly relevant to donor-acceptor systems where the K_2 terms for locally excited (LE) states are expected to be significantly larger than for charge transfer (CT) states. Specifically triplet LE states (^3LE) are stabilized this way as they are unaffected by exchange repulsion. Conversely, if HOMO/LUMO overlap is large, then the J_2 term associated to the singlet HOMO-LUMO transition may dominate pushing it above another singlet excited state. In principle, also the XC_2 term could dominate. However, we find that it generally only has a minor contribution in the examples presented here.

In summary, the discussion suggests that the post-MO terms are particularly important for LE states where triplet LE states are stabilized via K_2 and singlet LE states are destabilized via J_2 . The **ACRFLCN** and naphthalene molecules (see Fig. 1) are chosen to exemplify these two cases. We will discuss these molecules using the canonical MO picture, the energy component analysis described above, and a pictorial analysis scheme.

ACRFLCN: CT and LE states

ACRFLCN is a bichromophoric molecule comprised of an acridine donor and a triphenylamine acceptor. The chromophores are connected by a spiro junction forcing them into a perpendicular conformation reducing conformational flexibility and, thus, complexity providing a convenient example for this study. The frontier orbitals of **ACRFLCN** are shown in Fig. 3. The HOMO is located on the donor unit (right) whereas the LUMO, along with the HOMO-1 is on the acceptor unit (left). The HOMO and LUMO are associated to independent electron detachment and attachment processes. In line with expectations Fig. 3, thus, indicates that electron detachment involved in the oxidation of **ACRFLCN** would occur on the donor unit whereas electron attachment (reduction) occurs on the acceptor. Accordingly, the HOMO/LUMO transition is classified as an intramolecular CT state. By contrast, the HOMO-1 is located on the acceptor and the HOMO-1/LUMO transition is, thus, classified as LE.

TDDFT computations are now performed to study how the orbitals just described are involved in the lowest singlet and triplet states (see Table 2). The HOMO/LUMO gap, computed at the LRC- ω PBEh level is 5.856 eV whereas the gap between HOMO-1 and LUMO is 6.933 eV. Nonetheless, we find that the lowest excited state overall, the T_1 state, located at 3.030 eV, is reached predominantly *via* the HOMO-1 to LUMO transition forming a locally excited state, denoted henceforth ^3LE . It is followed by T_2 (^3CT) at 3.324 eV which is essentially a pure HOMO $\rightarrow$ LUMO transition. The S_1 (^1CT) state is found just 0.012 eV

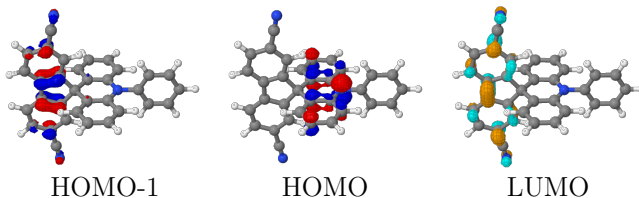


Figure 3: The HOMO-1, HOMO, and LUMO orbitals for ACRFLCN computed at the LRC- ω PBEh/def2-SV(P) level of theory

Table 2: Excitation energies (ΔE , eV), oscillator strengths (f), and leading configurations for the first two singlet and triplet states of ACRFLCN computed at the LRC- ω PBEh/def2-SV(P) (TDA) level of theory.

State	ΔE	f	Configurations ^a
³ LE (T_1)	3.030	-	$0.85h_1l + 0.37h_3l$
³ CT (T_2)	3.324	-	$0.99hl$
¹ CT (S_1)	3.336	0.000	$0.99hl$
¹ LE (S_2)	4.258	0.334	$0.91h_1l$

^a Dominant configurations and coefficients; h_xl_y refers to the HOMO- $x \rightarrow$ LUMO+ y transition.

higher in energy than T_2 with an almost identical electron configuration to the T_2 state. S_1 is a dark state with negligible oscillator strength. Finally, S_2 (¹LE) is a bright locally excited state at 4.258 eV and is reached *via* the HOMO-1 $\rightarrow$ LUMO transition.

Before continuing we note that the energetic difference between S_1 and T_1 as mentioned above is 0.306 eV, somewhat higher than the reported experimental gap of ≈ 0.1 eV.²⁵ The reason for this discrepancy is that the presented values are vertical excitation energies in gas phase whereas experiment considers structurally relaxed molecules in solution. We have studied the influence of structural relaxation on solvation on related systems in detail⁶⁰ but want to omit such a discussion here to focus on the presented decomposition for vertical gas-phase excitation energies. In the future, it is certainly possible to extend the presented analysis scheme to include contributions from solvation and structural relaxation.

Based on the data presented, we highlight two key points: first, the lowest excited states of each multiplicity are of different character (³LE and ¹CT) where only the singlet is formed via the HOMO/LUMO transition. Second, the energy difference between the ¹LE and ³LE states (1.228 eV) is significantly larger than that between the ¹CT and ³CT states (0.012 eV).

To rationalise these findings we proceed to the energy component analysis presented above. To develop an intuitive graphical representation for the energy contributions, we first realise that the one-electron terms h' and the exchange repulsion J_2 are always positive whereas the exciton binding K_2 is generally negative. Furthermore, we note that in the

presented LRC- ω PBEh computations XC_2 is always negative (but this varies with different functionals). We now plot these in a bar graph by stacking the positive contributions on top of each other and using the negative contributions as an offset. Using this representation, the total excitation energy amounts to the top of the left bar for each state. This analysis applied to **ACRFLCN** is presented in Fig. 4; the full raw data is provided in Table S1 of the ESI. The green bar in Fig. 4 represents h' and, on top of it, if applicable, we plot J_2 (blue). These terms are offset by the negative contributions K_2 and XC_2 shown on the right.

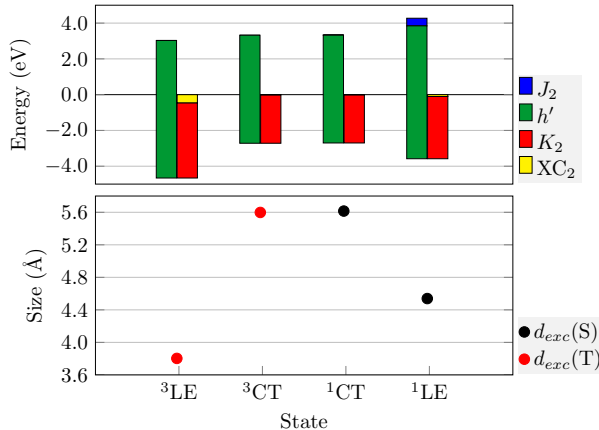


Figure 4: Energy components (top) and exciton size descriptors (bottom) for the computed excited states of ACRFLCN. Exciton sizes d_{exc} are plotted with singlets in black and triplets in red.

It is apparent that the h' term (green) for the ^3LE state of **ACRFLCN** (7.960 eV) is significantly larger than the analogous term for the ^3CT state (6.041 eV) in line with the latter representing the HOMO/LUMO transition. Nonetheless, ^3LE is the lowest state due to its enhanced K_2 (and in part XC_2) term. The difference in character between ^3LE and ^3CT is also reflected in the exciton size (d_{exc}), which is below 4.0 Å for the former and above 5.5 Å for the latter. More generally, we find an inverse correlation between the K_2 and d_{exc} values. Moving from ^3CT to ^1CT , we find that both states are almost indistinguishable in terms of their energy contributions as well as exciton size. In line with standard models,¹⁸ the exchange repulsion vanishes for a state with complete charge separation and this is reflected by an almost vanishing J_2 term of 0.010 eV.

As opposed to the ^1CT state, the ^1LE state has a distinctly different appearance when compared to its triplet counterpart showing enhanced exciton size d_{exc} (4.539 vs 3.803 Å) and less negative K_2 (-3.479 vs -4.204 eV). The obtained exchange repulsion (J_2) is 0.426 eV. If the ^3LE and ^1LE wavefunctions were equivalent aside from the spin-coupling, then their energy difference would be entirely determined by J_2 . However, this is clearly not the case here. The gap between ^3LE and ^1LE is 1.228 eV; out of these only +0.426 eV are due to J_2 whereas differences in the Coulomb attraction between the two states ($\Delta K_2 = 0.725$ eV) nominally make a bigger contribution; changes with the one-electron ($\Delta h' = -0.262$ eV) and exchange-correlation ($\Delta XC_2 = 0.353$ eV) terms cancel out approximately. To understand this discrepancy, we compute the formal J_2 term obtained if the ^3LE wavefunction were directly projected into the singlet manifold. Strikingly, we obtain a formal J_2 term of 2.784 eV (more than twice the gap between ^1LE and ^3LE) putting the excitation energy of the associated singlet state to 5.814 eV. We can now interpret the change in the ^1LE wavefunctions compared to ^3LE as an “attempt” of the ^1LE state to lower its energy by minimising the exchange repulsion. More accurately, by virtue of being higher in energy the singlet HOMO/LUMO configuration interacts more strongly with other electronic configurations. This will be discussed in more detail below.

To provide a graphical interpretation of the two-electron terms we use a formalism developed in Ref. 24. This is exemplified for the S_1 , S_2 , and T_1 states in Fig. 5. The T_2 , which closely resembles S_1 is presented in the ESI (Fig. S5). The *hole* and *electron* NTOs (cyan and orange) are shown at the center. The associated *electron* (blue) and *hole* (red) densities and their ESPs, related to K_2 , are shown above. Transition densities and their ESPs, giving rise to J_2 are shown below.

Starting with the top of Fig. 5, we first regard the *hole* and *electron* densities. These are formed as weighted sums over the NTOs and, since only one NTO pair contributes significantly to either state, these densities closely reflect the corresponding NTOs. Considering that only one NTO pair contributes, it is appropriate to apply Eq. (14), i.e. the K_2 term can

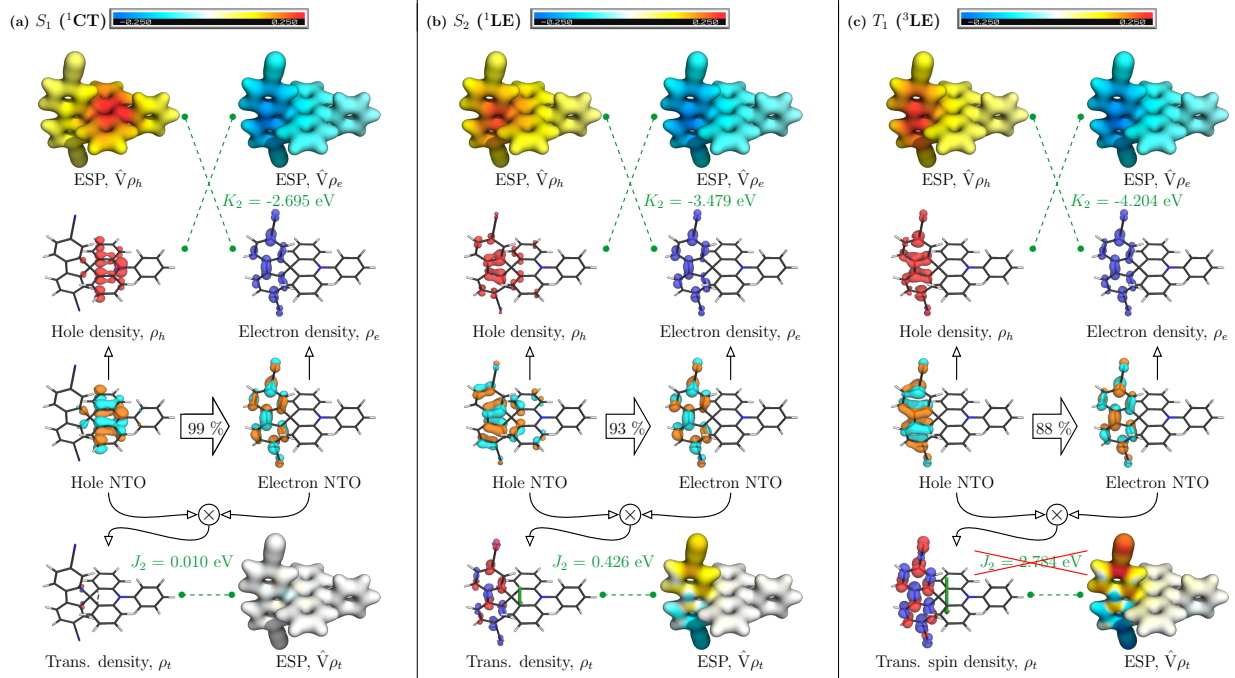


Figure 5: Analysis of the energetic contributions to the (a) S_1 (^1CT), (b) S_2 (^1LE), and (c) T_1 (^3LE) states of ACRFLCN. The electron and hole densities, and the ESPs they induce are shown. The Coulomb interaction between hole/electron density and electron/hole ESP are shown as a green dashed line. The contributions of the dominant natural transition orbitals are shown in the centre. The transition density and its ESP are shown at the bottom. Isovalues used: 0.05 for orbitals, 0.004 for densities, 0.05 for the ESP map.

be interpreted as a static attraction between the *electron* and *hole* densities. To visualise this term, we compute the electrostatic potentials (ESP) induced by the *electron* and *hole* densities. The *electron* density and its ESP possess a very similar shape for all three states shown, being concentrated on the acceptor group. However, a clear difference is observed for the *hole* density, which is located on the donor group for S_1 but on the acceptor group for the other states. Crucially, the difference is strong enough to produce a markedly different appearance of the ESP of the *hole* density (denoted $\hat{V}_{\rho_h}$ in Fig. 5). The ESP has its most positive contribution (red) on the donor for S_1 with a notable decrease on the acceptor group and vice versa for the CT states. The approximate K_2 term is now obtained as the term $\langle \rho_e | \hat{V} | \rho_h \rangle$, that is, by computing the overlap of the *hole* ESP with the *electron* density or vice versa. Clearly ρ_e lies in an area of lower *hole* ESP for the S_1 state than it does for the other states and this explains the reduced K_2 term.

Proceeding to the bottom part of Fig. 5, we can now discuss the emergence of the J_2 term via the transition density. The product of the hole and electron NTOs leads to the transition density for the excited state (following Eq. (9) and setting $r_e = r_h$). The computed transition density is shown in blue and red at the bottom of Fig. 5. Following Eq. (8), the J_2 term is given as the self-repulsion of the transition density. In analogy to the Coulomb term we visualise it via its ESP and the J_2 term is indicated via the green dotted lines in the lower half of the figure. For the ^1CT (S_1) state, we find that the transition density and its ESP almost vanish completely everywhere in space, explaining its small J_2 term. Conversely, notable contributions for the transition density are found for the ^1LE (S_2) state. The transition density induces a strong ESP with a clear dipolar shape, explaining the stronger J_2 term. Note, that the dipolar form of the transition density also induces the strong transition dipole moment, shown as a green arrow in Fig. 5.

Finally, we revisit the question of the differences in wave functions between the ^1LE (S_2) and ^3LE (T_1) states. Both of these are formally composed of the HOMO-1/LUMO transition. However, clear differences in wave functions are apparent in Fig. 5. Most notably, the *hole* NTO for the T_1 state (Fig. 5 (c)) is entirely localized on the acceptor unit whereas notable admixture of donor contributions is found for S_2 . This difference leads to enhanced charge transfer as seen from the enhanced d_{exc} value of S_2 vs T_1 in Fig. 4. Importantly, this causes reduced electron-hole overlap and hence a diminished transition density and ESP causing the reduction in the J_2 term. In addition, the delocalization of the hole also leads to more delocalization of its ESP $\hat{V}\rho_h$ toward the donor and hence the reduced K_2 term. A more detailed depiction of the natural transition orbitals involved is shown in Figs S1 and S4. Without going into too much detail here, we want to stress that the NTOs are clearly different between singlet and triplet states in terms of their shapes and occupations, highlighting again the differences in the underlying wave functions. In summary, the S_2 state is differentiated from the T_1 state through enhanced charge transfer character, which reduces exchange repulsion (via J_2) but also diminishes exciton binding effects (via K_2). Similar

differences between singlets and triplets were also observed in related push-pull systems.⁶⁰

An alternative viewpoint on these phenomena has been presented in Ref. 61 highlighting that due to correlation the singlet-triplet gaps are often only about half the value that would be expected from the HOMO/LUMO exchange integral. Furthermore, it is interesting to view these results in the context of unrestricted (UKS) and restricted open-shell (ROKS) Kohn-Sham computations on singlet excited states.⁶²⁻⁶⁴ A spin-broken UKS state only experiences half the J_2 term compared to the spin-adapted singlet of the TOTEM shown in Eq. (5) meaning that UKS should severely underestimate singlet state energies. However, we have just shown that correlation and relaxation effects also lead to halving of the J_2 term meaning that, in summary, spin-broken UKS is a surprisingly good approximation to singlet excited states.

In summary, we find that the locally excited triplet state of **ACRFLCN** lies below the HOMO/LUMO transition (3CT) due to its enhanced Coulomb binding term (K_2). By contrast, the exchange repulsion (J_2) dominates for the singlets, keeping the singlet HOMO/LUMO transition (1CT) the lowest singlet state.

Naphthalene: ionic and covalent states

Naphthalene is a paradigmatic case of an aromatic molecule, possessing a number of rather closely spaced $\pi\pi^*$ states with distinct characteristics despite being all constructed from the same set of MOs. Indeed, discussion of the MOs alone is by far not sufficient to characterize these states and a number of different viewpoints, such as Platt’s nomenclature⁶⁵ and Pariser’s $+/-$ nomenclature³⁰ have been introduced to obtain a clearer picture.³⁶ The critical realization in this case is that not only the MO transitions as such are of relevance but that also the relative signs between the different interacting configurations are decisive.^{17,27,28} The effect of this is seen in physically observable quantities such as oscillator strengths and energies but also resurfaces in the computational description posing severe and sometimes unexpected challenges.^{32,33}

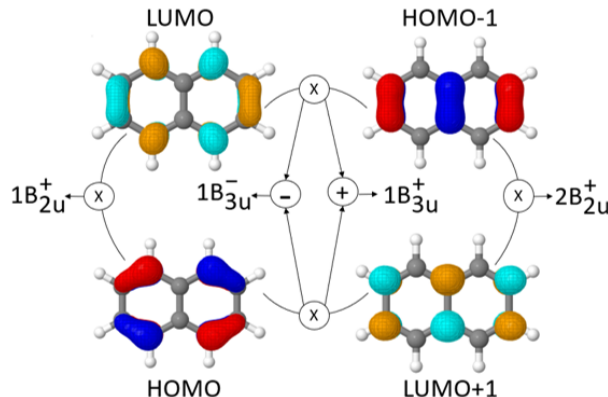


Figure 6: The linear combinations of canonical orbitals leading to the first four singlet and triplet excited states of naphthalene according to the dominant configuration.

In the following, we analyse the lowest four singlet and triplet excited states of naphthalene. All these states can be explained by transitions from the HOMO-1 and HOMO to the LUMO and LUMO+1, which are presented graphically in Fig. 6. The HOMO/LUMO and HOMO-1/LUMO+1 transitions mix only weakly and yield two individual states of B_{2u} symmetry, denoted $1B_{2u}^+$ and $2B_{2u}^+$, as shown to the left and right. Conversely, strong mixing is observed between the HOMO/LUMO+1 and HOMO-1/LUMO transitions yielding two states ($1B_{3u}^-$ and $1B_{3u}^+$) distinguished only by the sign used in their linear combination. Note that the signs in the linear combination depend on the orbital signs and we use the conventions introduced by Pariser.³⁰ Furthermore, it is important to realise that the singlet “+” states are interpreted as ionic states and the singlet “-” states as covalent states within valence-bond theory.³¹ The four possible states formed are shown in the middle line of Fig. 6; coupling them with singlet or triplet spin yields the eight states of interest.

Computational data on the states is presented in Tab. 3. Here, we present excitation energies, oscillator strengths and configurations along with the LOC descriptor (see Eq. (17)). The LOC descriptor provides a convenient means to differentiate between “+” and “-” states where only the former possess non-vanishing LOC values. The lowest excited state is $1^3B_{2u}^+$ (or 3L_a) at 3.071 eV, reached *via* the HOMO/LUMO transition. Three more triplet states ($1^3B_{3u}^+$, $1^3B_{3u}^-$ and $2^3B_{2u}^+$) follow in the range between 4.212 and 4.684 eV. The first singlet

Table 3: Excitation energies (ΔE , eV), oscillator strengths (f), leading configurations and LOC descriptor computed for the first four singlet and triplet states of naphthalene at the LRC- ω PBEh/def2-SV(P) (TDA) level of theory

State	ΔE	f	Configurations ^a	LOC
$1^3B_{2u}^+(T_1)$	3.071	-	$0.93hl$	0.138
$1^3B_{3u}^+(T_2)$	4.212	-	$0.74h_1l + 0.64hl_1$	0.063
$1^3B_{3u}^-(T_3)$	4.510	-	$0.74hl_1 - 0.65h_1l$	0.000
$2^3B_{2u}^+(T_4)$	4.684	-	$0.95h_1l_1$	0.088
$1^1B_{3u}^-(S_1)$	4.779	0.000	$0.70hl_1 - 0.70h_1l$	0.000
$1^1B_{2u}^+(S_2)$	5.085	0.097	$0.93hl$	0.074
$1^1B_{3u}^+(S_3)$	6.815	2.023	$0.68h_1l + 0.68hl_1$	0.040
$2^1B_{2u}^+(S_4)$	6.990	0.393	$0.89h_1l_1$	0.072

^a Dominant configurations and coefficients; h_xl_y refers to the HOMO- $x \rightarrow$ LUMO+ y transition.

state ($1^1B_{3u}^-$, also known as 1L_b) is dark and lies at 4.779 eV. Interestingly, S_1 possesses the same character as T_3 highlighting the different energetic contributions that act on singlet and triplet states leading to different ordering of the states in the respective spin manifolds. The second singlet state ($1^1B_{2u}^+$ or 1L_a), finally, is reached via the HOMO/LUMO transition in analogy to T_1 . With an oscillator strength of 0.097 it forms the first notable band in the absorption spectrum of naphthalene. Two more bright states ($1^1B_{3u}^+$ and $2^1B_{2u}^+$) follow near 7 eV. The state ordering, i.e., the dark 1L_b state below the bright HOMO/LUMO transition (1L_a) followed by even brighter high-energy states, is well-documented in the literature^{27,66} and we shall explain this energetic ordering in the following.

Tab. 3 illustrates the importance of post-MO contributions in the analysis of excitation energies. The states presented span a range of almost 4 eV despite being composed of the same four MOs. It is particularly noteworthy that T_1 is separated from all the other states by more than 1 eV while S_3 and S_4 are separated from the others by almost 2 eV. Strikingly, there is a difference of more than 2.5 eV between the $1^3B_{3u}^+$ and $1^1B_{3u}^+$ states, which are constructed via analogous orbital transitions and only differ in electron spin. We will now endeavour to explain these differences using the energy component analysis for naphthalene, as presented in Fig. 7.

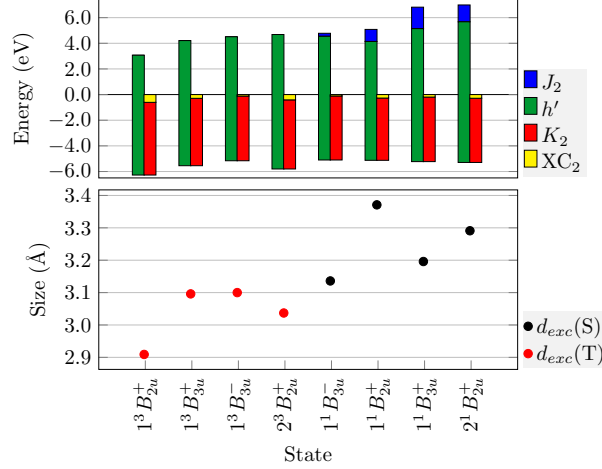


Figure 7: Energy components and exciton sizes for the computed excited states of naphthalene. Exciton sizes d_{exc} are plotted with singlets in black and triplets in red.

Unlike **ACRFLCN** all of the excited states in naphthalene can be considered locally excited as seen by their low exciton sizes ($d_{exc} < 3.5$ Å). As a consequence, all K_2 values are fairly similar (around -5 eV) with no dramatic changes as seen in Fig. 4. Nonetheless, the balance between the different terms has a crucial effect. We first investigate why T_1 ($1^3B_{2u}^+$) lies more than 1 eV below T_2 ($1^3B_{3u}^+$). Only 0.413 eV are due to the orbital energies ($\Delta h'$). Additional important contributions come from Coulomb binding ($\Delta K_2 = 0.414$ eV) and the response of the exchange-correlation potential ($\Delta XC_2 = 0.313$ eV). These indicate a more local nature for T_1 , which is also reflected in its smaller exciton size ($d_{exc} = 2.909$ Å) and higher LOC value.

Next, we may ask why the $1^3B_{3u}^-$ state lies above the $1^3B_{3u}^+$ state by about 0.3 eV. Being composed of the same orbital transitions, both states have almost equivalent one-electron components ($\Delta h' = -0.088$ eV). Conversely, the difference lies in exciton binding ($\Delta K_2 = 0.209$ eV) and exchange-correlation ($\Delta XC_2 = 0.178$ eV). These values are enhanced due to the more local nature of the $1^3B_{3u}^+$ state as indicated by the *LOC* descriptor in Table 3. The next state, $2^3B_{2u}^+$, is the HOMO-1/LUMO+1 transition. It lies above the $1^3B_{3u}^-$ state on account of the increased one electron component ($\Delta h' = 0.816$ eV), consistent with the fact that the canonical orbitals involved (HOMO-1/LUMO+1) are further apart energetically.

This increase is offset only in part by the two-electron terms.

The first singlet state (S_1 , $1^1B_{3u}^-$) is the counterpart to the T_3 state. We find that their energy components are almost identical, the only major difference being the J_2 term of 0.234 eV, which is responsible for the energetic difference between the two states.

The HOMO/LUMO singlet state ($1^1B_{2u}^+$) follows as S_2 , being the sixth excited state overall. Its one-electron component of $h' = 9.270$ eV is lower than the four preceding states in line with being the HOMO/LUMO state. However, it is pushed up in energy due to its exchange repulsion term of $J_2 = 0.939$ eV. In addition, when compared to the corresponding triplet state (T_1), we find that binding via K_2 and XC_2 is strongly reduced. Indeed, the gap between S_2 and T_1 is 2.014 eV meaning that only half of this can be explained via the J_2 term where the other two-electron terms account for the rest. Conversely, the formal J_2 term evaluated for T_1 is 4.203 eV. Thus, we are left with the same result seen for the ^1LE and ^3LE states of **ACRFLCN** where the formal J_2 term is strongly reduced when going from the triplet to the singlet, but this is done at the cost of other two-electron terms.

Next, the $1^1B_{3u}^+$ state is found as S_3 lying 1.730 eV higher than the previous states. This state is dominated by the HOMO-1/LUMO and HOMO/LUMO+1 transitions just as T_2 , T_3 , and S_1 . However, its energy components differ markedly from these states. The one-electron components ($h' = 10.373$ eV) are significantly higher and a prominent exchange term ($J_2 = 1.672$ eV) is observed, both of which contribute to the higher excitation energy. Finally, the $2^1B_{2u}^+$ state is obtained as S_4 possessing an enhanced one-electron gap ($h' = 10.971$ eV) due to being the HOMO-1/LUMO+1 transition along with a substantial exchange repulsion ($J_2 = 1.313$ eV).

To understand the observed trends in some more detail, we plot the transition densities and their ESPs, in analogy to the bottom part of Fig. 5. These are shown in Fig. 8 for the B_{3u} states all of which are even mixtures of the h_1l and hl_1 configurations, only differentiated by the relative signs. Analogous plots for the B_{2u} states are presented in Fig. S14. First, we observe that the transition densities of all the “+” states are concentrated on the atoms

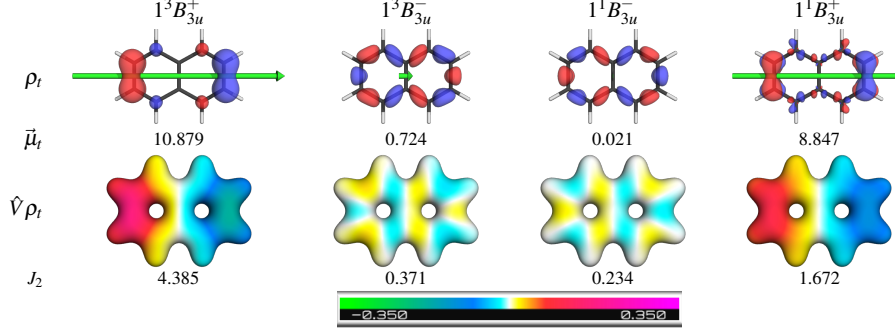


Figure 8: The transition densities (ρ_t), transition dipole moments ($\vec{\mu}_t$) in Debye, ESPs ($\hat{V}\rho_t$), and computed exchange repulsion components (J_2) in eV for the B_{3u} states of naphthalene. The ESP colour coding is shown. The transition dipole moment ($\vec{\mu}_t$) is shown as a green arrow of length $2\vec{\mu}_{0I}/e$. Isovalues used: 0.05 for orbitals, 0.004 for densities, 0.05 for the ESP map.

whereas they are between them for the “−” states.^{29,36} This is in agreement with the LOC descriptors: The “+” states show enhanced LOC and hence *electron* and *hole* are on the same basis function on a given atom, producing transition density on that atom. For the “−” states, on the other hand, the transition density is only formed via “differential overlap” terms, i.e., as the overlap of basis functions on different atoms (see also Ref. 67). As a consequence, the transition moments and ESPs are significantly reduced for the “−” states when compared to the “+” states, which is reflected in the oscillator strengths and energies of the singlet states shown in Tab. 3. We find a general positive correlation between J_2 and the transition moments explaining why brighter states are pushed up in energy (cf. Table 3).

Finally, we note the differences in the shape of the transition density for the singlet and triplet B_{3u}^+ states. The triplet state only shows π -contributions on the outer atom. Interestingly, for the singlet there are also some σ contributions on all the atoms involved. These σ contributions serve in lowering the J_2 term,²⁴ albeit at the cost of raising h' and K_2 as discussed above. These σ contributions are also apparent in the NTO representation (Figs S12, S13) where they appear as $\sigma\sigma^*$ excitations with collective weights of a few percent.

In summary, we note that the presented analysis scheme provided a rationale for the energy ordering of all eight states considered. In particular, we could explain why the

HOMO/LUMO transition does not form the lowest singlet state by considering its enhanced J_2 term. The discussion also showed that the wavefunctions involved are more complicated than one might presume at first. In particular, mixing of different configurations is involved lowering the J_2 terms of the ionic singlet states. This discussion also highlights why the computational description of the related ionic and covalent states can be highly challenging and, specifically, why correlation of the σ system is decisive to obtain accurate energies.^{48–50}

Conclusions and Outlook

Within this work, we presented an energy component analysis for excited states, elucidated the physical meaning of the different terms and used it to answer one perplexing question: why is the lowest excited state of a given spin-multiplicity not always derived from the HOMO/LUMO transition? It was shown that excited state energies can be understood as a sum of a weighted orbital energy difference (h'), a Coulomb attraction term (K_2), an exchange repulsion term (J_2), and additional two-electron exchange/correlation terms (XC_2). The Coulomb attraction term favours locally excited states, in particular triplets. The exchange repulsion term, on the other hand, disfavors locally excited singlet states. Therefore two cases can be constructed where the HOMO/LUMO transition is not the lowest state: (i) if the HOMO/LUMO transition is of CT character, then a locally excited triplet may be of lower energy; (ii) if the HOMO/LUMO transition is locally excited, then a different singlet state may be lower by minimising exchange repulsion (either through its CT character or interaction of different electronic configurations). To exemplify case (i), we used the push-pull system **ACRFLCN** while the naphthalene molecule was used for case (ii).

For **ACRFLCN**, we found indeed that the HOMO/LUMO transition formed the T_2 state. This was explained by the fact that the HOMO-1/LUMO transition represented a locally excited state with enhanced Coulomb attraction, which pushed it below T_2 despite

its larger MO energy gap. The order was reversed for the singlets due to enhanced exchange repulsion in the locally excited state, compensating for Coulomb attraction.

It was found that for naphthalene the T_1 state is reached *via* the HOMO/LUMO transition whereas S_1 is reached by a linear combination of the HOMO-1/LUMO and HOMO/LUMO+1 transitions forming the $L_b/1^1B_{3u}^-$ state. The eight computed excited states for naphthalene were all found to be reached via some linear combination of transitions which involve only four canonical orbitals: the HOMO-1, HOMO, LUMO and LUMO+1. We have shown how the presented energy component analysis scheme offers a rationale for the ordering of all these states.

In a more general sense, we have found in agreement with previous work⁶¹ that the simple picture of equating the singlet-triplet gap to twice the HOMO/LUMO exchange integral is not quantitatively accurate. Evaluating the formal exchange repulsion using the triplet states yielded about twice the singlet-triplet gap actually obtained. Conversely, the singlet states lower their energy via additional orbital relaxation and correlation terms redistributing the energy penalty from J_2 into the h' and K_2 terms. These observations provide new insight into the challenges of providing a balanced computational description for molecular singlet and triplet states of ionic or covalent character, which have been observed for single^{33,68,69} and multireference methods.^{48–50} A deeper understanding of these issues could ultimately help in the development of targeted corrections to these methods.

We have presented our analysis protocol for TDDFT being a convenient and widely used method. However, we want to stress that the general physics involved is independent of the computational protocol. It is certainly also possible to use this type of analysis in a correlated ab initio framework. For example in the context of the algebraic diagrammatic construction (ADC) method, the h' term could be represented via zeroth order energies, J_2 and K_2 via first order, and XC_2 would be made up of higher order terms. Moreover, many of the concepts mentioned are naturally included in the GW framework.^{45,46} Alternatively, it is possible to separate the analysis from any quantum chemistry method using Dyson orbitals

and many-body ionisation potentials/electron affinities as a starting point. Therefore, we hope that the presented analysis will be useful for, both, practical calculations and formal discussions.

Acknowledgement

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Supporting Information Available

Numerical values of energy components (Table S1); NTOs and ESP analysis of ACRFLCN (Figs S1-S5); NTOs and ESP analysis of naphthalene (Figs S6-S14).

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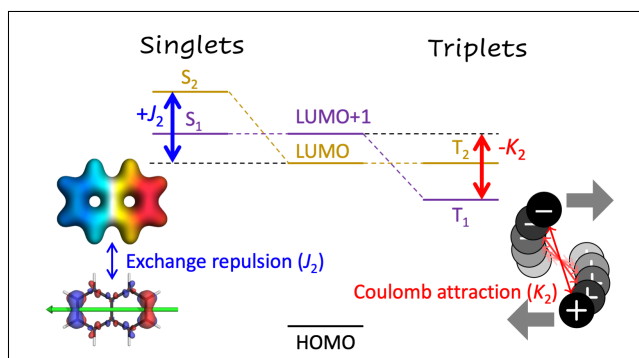
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TOC Graphic



A scheme for decomposing energy components is proposed. It is shown how the exchange repulsion (J_2) destabilises singlets whereas the Coulomb attraction (K_2) can strongly stabilize triplets, both interactions may affect the state ordering making the HOMO/LUMO transition the second excited state.